

Description of Information System

- The Description must **not exceed 5 pages**, using A4-sized paper in portrait orientation
- Margins: Top: 2cm Bottom: 2cm Left: 2cm Right: 2cm Gutter: 1cm
- Font size: 12 pt
- The description must contain the following information:

Application No.: (To be filled in by Contest Committee)

Title of Information System: ALL Wheelchair: AI Motion Tracking System for Monitoring Health and Activity for Wheelchair Users.

University / Institution: Thammasat University

Department / Faculty: Faculty of Allied Health Sciences, Faculty of Engineering, and Sirindhorn International Institute of Technology

I. Preface

Wheelchair users often struggle to stay active due to a lack of accessible exercise options and support. This inactivity can cause physical health risks and mental health issues. Meanwhile, current solutions do not meet their needs or help them maintain good health. We, therefore, provide a new solution “ALL Wheelchair: AI Motion Tracking System for Monitoring Health and Activity for Wheelchair Users” to make exercise accessible and enjoyable for wheelchair users.

Compared to 2D motion analysis, the developed motion tracking system yields an acceptable error rate (<1%) and provides high validation of wheelchair tracking ($r > 0.90$) for traveling distance, velocity, and time.

- Implemented with the Wheelchair Basketball team of Thammasat University
- Thailand Patent no. 21587
- Applications for Thailand Patent no.2303002152, 2401001717 & 2401002438

II. Innovation Description

Nowadays, many people use wheelchairs as a device that helps facilitate mobility for users with various mobility limitations, such as those who are with disabilities and the frail elderly. Moreover, There're 703 million elderly worldwide (aged 65 years and older). The number is expected to increase significantly in the coming decades due to aging populations in many countries. Therefore, the ratio of frail elderly who need to use wheelchairs will tend to increase as well as disabled persons. According to the World Health Organization (WHO), there are more than 131 million wheelchair users worldwide which is also a huge number.

Wheelchair users are twice the risk of developing health conditions and may face many struggles in society in their daily lives due to limited ability. Our project “ALL Wheelchair” mainly focuses on monitoring health and activities including exercising purposes for wheelchair users. Our project is developed from the “YES Wheelchair” project which analyzes the motion of wheelchair users by using the M5StickC sensor to detect the motion and send the data to the monitor using Bluetooth classic. The real-time data displayed on the

monitor consists of accelerometer and gyroscope readings in the x, y, and z axes. Then we developed a new version called “ALL Wheelchair” which acts as a smartwatch and Wii Mote for wheelchair users. We also developed specific games for specific groups using sensors on the wheelchair’s wheels and on the user’s wrist to monitor activities and vital signs such as heart rate and oxygen saturation, including control the games. Currently, we changed the sensor from M5StickC to XIAO nRF52840 due to the battery capacity.

The developed ALL Wheelchair system proved to be a significantly high reliability and validity system compared to a 2D motion analysis with $r > 0.90$ and an acceptable level of percentage error $< 1\%$ in a study conducted on 30 participants who performed 4 skill-related fitness tests for wheelchair users.

Our project was inspired by Nintendo Wii which uses Wii Mote to detect activities. So, we use the sensor which attaches on both wheel and user wrist as Wii Mote to detect wheelchair and user’s motion and change those motions to control in-game. We also have created more than 20 specific games such as Astro Avenger, Ball Game, and Flappy Bird which have different methods to control the character. Each game aims for different physical measurement outcomes and is designed for different target groups such as wheelchair athletes, frail elderly, etc.

Also, the fall detection system will notify the caregiver when a wheelchair user falls. Moreover, we have developed a Pre-Cautions system to predict movement of users by applying AI to learn the behavior of wheelchair users for independence going outside by themselves.

Another important feature is called “Smart Wheelchair”, which we integrate with the smart home system. The wheelchair will be able to control various functions and integration with home devices. Therefore, it will be applied as a smart innovation in allowing wheelchair users to communicate with the smart house itself.

Targets

- Wheelchair users
- Healthcare providers
- Caregivers

User Needs

Wheelchair users also want to live a healthy and happy life. However, they cannot reach their goals due to the obstacles of their limitation. Nowadays, there is a lack of suitable tracking and monitoring activity devices. Our system addresses their needs by providing an accessible, engaging, and effective means to persuade them to have more daily activities and exercises through motion-controlled gaming. It doesn’t only promote physical activity but also enhances their mental well-being.

User Consultation

Extensive consultation with 30 wheelchair users aged between 18 – 70 years old and 15 healthcare providers and caregivers has been suggested to ensure the system meets the specific needs and preferences of the target users.

III. System Functions

The ALL-Wheelchair system operates through a range of integrated functions, designed to track, analyze, and enhance the user’s movement and health while providing critical safety measures. These technical operations work behind the scenes to ensure seamless interaction

between the user, their wheelchair, and their environment.

1. Sensor Data Acquisition and Processing: The system gathers real-time data from embedded sensors on the wheelchair and wristband. This data is wirelessly transmitted to the central processing unit for analysis.

2. Real-Time Motion Analysis: Continuous monitoring of linear acceleration (x, y, z axes) and angular velocity enables real-time tracking of the wheelchair's movement, allowing for detailed analysis of speed, distance, and user behaviour patterns.

3. Health Data Monitoring: The system collects and analyses vital signs such as heart rate and oxygen saturation, ensuring accurate real-time monitoring of the user's physical health.

4. AI-Based Fall Detection: The system's fall detection algorithm processes both motion data and predictive AI analysis to detect potential falls which immediately alert caregivers in case of detected abnormalities.

5. Gaming Input Integration: User input from motion sensors is used to control interactive games, transforming physical movements into gameplay actions.

6. Bluetooth and Wireless Data Transmission: Communication between the wheelchair sensors, user wristband, and central processor occurs wirelessly via Bluetooth. The system ensures smooth transmission and low-latency data handling.

7. Data Storage and Analytics: Data collected from the wheelchair and sensors is stored in a database (Firebase and SQLite), where it can be accessed for comprehensive analysis.

8. System Integration with Smart Home Devices: The system's processing unit interfaces with smart home devices through Bluetooth or IoT protocols

9. Real-Time Alerts and Notifications: Alerts generated by the system, such as fall notifications or health anomalies, are sent via LINE Notify to caregivers in real-time.

10. Mobile and Cloud Synchronization: The system synchronizes user data with mobile applications and cloud-based platforms, enabling caregivers and users to access real-time updates and comprehensive reports from any location.

IV. System Features

The ALL-Wheelchair system offers the following user-facing features, designed to provide wheelchair users with a safe, interactive, and health-focused experience:

1. Real-Time Motion Tracking: Users can monitor their wheelchair's movements in real-time, viewing data such as speed, distance traveled, and activity duration through a user-friendly interface.

2. Health Monitoring Dashboard: A comprehensive health dashboard displays vital

signs, including heart rate and oxygen saturation, providing users and caregivers with easy-to-read health metrics and trends over time.

3. Interactive Gaming for Physical Exercise: The system offers engaging motion-controlled games that encourage users to exercise by using the wheelchair's motion as gameplay input.

4. Fall Detection and Safety Alerts: If the system detects a fall or abnormal movement, it sends an instant notification to both the user and their caregiver. The AI also predicts risky behavior, offering proactive safety measures.

5. Smart Home Integration: Users can control home devices like lights, doors, and air conditioning directly from their wheelchairs via voice commands or motion inputs, enhancing independence and comfort.

6. Comprehensive Data Reports: Users and caregivers can view historical data and progress reports that detail mobility trends, activity patterns, and health metrics. These reports provide insights into the user's long-term health and physical activity performance.

V. System Development Tools and Techniques

Hardware: Sensors, such as accelerometers and gyroscopes, embedded in wheelchairs and wristbands to capture motion data.

Software Tools:

- 1. Pygame** for motion-controlled gaming.
- 2. PyAutoGUI** for user interaction automation.
- 3. Firebase, SQLite, and Streamlit** for data storage, processing, and live visualization.
- 4. Grafana and Node-RED** for dashboard creation and monitoring of real-time analytics.
- 5. InfluxDB** for time-series data storage and analysis.
- 6. LINE Notify** for notifications and emergency alerts.

AI Techniques:

- 1. AI models, including Long Short-Term Memory (LSTM) networks,** are used to analyze sensor data for fall detection and risk assessment.
- 2. Data processing algorithms** determine risk thresholds using calculated magnitudes from acceleration and angular velocity data.

VI. System Users

The system is designed to cater to a broad range of users, including:

1. **Wheelchair Users:** The primary beneficiaries, with a specific focus on improving health, mobility, and safety.
2. **Healthcare Providers:** Medical professionals and therapists who can monitor user progress and offer personalized care plans.
3. **Caregivers:** Family members or personal caregivers who can monitor real-time health status and receive alerts in the event of falls or irregular activities.
4. **Researchers and Developers:** Those interested in studying mobility in wheelchair users and improving assistive technology.

VII. System Environment

The system operates in a comprehensive environment involving both hardware and software components:

1. **Physical Environment:** Sensors are attached to wheelchairs and wristbands, providing a non-invasive solution for tracking user movements and health parameters.
2. **Software Environment:** The system employs a suite of software tools for data capture, analysis, and visualization, including Bluetooth for data transmission, Firebase for cloud-based data management, and InfluxDB for real-time monitoring.
3. **Networked Infrastructure:** The system integrates seamlessly with mobile applications, allowing users, caregivers, and healthcare professionals to access information on smartphones. The LINE app is used for real-time notifications, ensuring continuous monitoring and communication.
4. **AI and Machine Learning:** The AI algorithms run in the background, analyzing movement data for risk assessments and fall detection, offering proactive alerts and warnings.

VIII. Conclusion

The "ALL Wheelchair: AI Motion Tracking System for Monitoring Health and Activity for Wheelchair Users" is a transformative solution that addresses a critical gap in accessible exercise and health monitoring for wheelchair users. Through the integration of advanced sensor technology, AI-driven fall detection, and real-time data analytics, the system promotes both physical health and independence. The inclusion of gaming and smart home integration further enhances user's health engagement and quality of life.

"We believe that having faith in the power of change is a brave act. Everyone has an ability to shape their own life. We're proud to be part of this change through this innovation"